

Roll No.

Total No. of Questions : 6]

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B. Tech. 4th Semester (New Scheme)

Examination, 2023-24

Microprocessor & Microcontroller

Paper - EL-405

Time : 3 Hours]

[Maximum Marks : 60

Note : - First question is compulsory. Attempt any two parts from remaining.

1. Write short answer :

5×2=10

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(1)

P.T.O.

- (i) What is the function of RIM and SIM instructions in 8085?
 - (ii) What is minimum and maximum mode of operation in 8086?
 - (iii) Give features of 80386 Processor.
 - (iv) What the function of IC8212?
 - (v) Give format and function of PSWR register in 8051.
2. (a) Draw and explain internal architecture of 8085. ✓
- (b) Draw Timing Diagram for the instruction MVI C, 32H and explain all the machine cycles present.
- (c) Write a Program for 8085, to add 10 bytes of data, stored in memory location starting at 7000 H. Store the Sum and Carry at 8001H & 8002H locations respectively.
3. (a) Explain the function of the following Pins of IC 8086 :
- (i) BHE'/S7
 - (ii) RQ'/GT'

- (iii) TEST' and LOCK'
 - (iv) DEN' & DT/R'
 - (v) HOLD & HLDA
- (b) Discuss different addressing mode of 8086 with suitable example.
- (c) Describe memory management in 8086? Explain.
4. (a) Write the control word for IC 8155 and explain different modes of timer operation.
- (b) Draw the block diagram of IC8259 and explain the function of all registers.
- (c) Discuss different modes of timer operation with the help of waveform in IC8254.
5. Write short note on any two of the following :
- (i) Pentium Super Scalar architecture
 - (ii) 80486 Processor
 - (iii) Alpha AXP processor

P.T.O.

(3)

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6. (a) Draw and Explain the block diagram of 8051. ✓
- (b) Describe internal RAM architecture of 8051.
- (c) Explain timer operation of 8051 and related SFRs.
